

Rationality of the center of a generic division algebra with a group action over a field of characteristic 0

Zhengyao Wu

June 17, 2026

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Abstract

Let F be a field of characteristic 0 and let $H = C_3 \times C_3$ act regularly as a transitive subgroup of S_9 . We resolve Problem 5.2 of Auel–Brussel–Garibaldi–Vishne: $Z_H(F, 9)$, the center of the generic division

algebra of degree 9 with H -action, is *not* stably rational and *not* rational, but is retract rational over F . The non-stable-rationality is proved by computing $H^2(H, M|_H) \cong C_9$ for the restricted Procesi lattice and showing that its exponent 9 is incompatible with stable permutation (all permutation lattices have H^2 -exponent dividing 3). This integral cohomological obstruction is of a different nature from the classical Bogomolov multiplier ($B_0(C_3 \times C_3) \cong C_3$), which governs the Noether problem but not $Z_H(F, 9)$. The proof generalizes to all odd primes p : for $H = C_p \times C_p \subset S_{p^2}$, one obtains $H^2(H, M|_H) \cong C_{p^2}$ while permutation lattices have exponent dividing p , so $Z_H(F, p^2)$ is not stably rational.

1 Introduction

Let F be a field of characteristic 0. For an integer $n \geq 1$ and a subgroup $H \subseteq S_n$, the *generic division algebra with H -action*, denoted $UD(F, n, H)$, has center $Z_H(F, n)$. Problem 5.2 of Auel–Brussel–Garibaldi–Vishne [?, Problem 5.2] asks:

Determine whether $Z_H(F, 9)$ is rational, stably rational, or retract rational over F , where $H = C_3 \times C_3 \subset S_9$ is a transitive subgroup.

The rationality problem for fields of the form $F(M)^G$ has a long history. Endo and Miyata [?, §1] developed the theory of quasi-permutation modules, providing the algebraic foundation for the rationality problem. Building on this, Endo–Miyata [?, Theorem 1.2] proved that $F(M)^G$ is stably rational over F if and only if M is a quasi-permutation $\mathbb{Z}[G]$ -lattice, where “quasi-rational” in the original terminology of [?] is synonymous with stably rational. Voskresenskii [?, Chapter 2] gave a comprehensive treatment via the birational geometry of algebraic tori. Colliot-Thélène and Sansuc [?, §1] developed flasque resolutions and their applications to rationality. Saltman [?, Section 3] introduced retract rationality and constructed the first examples of non-rational stably rational fields. Hoshi, Kang, and Yamasaki [?] systematically studied Noether’s problem and multiplicative invariant fields for small groups via unramified Brauer groups and Bogomolov multipliers [?, §4, Definition 1]. The H^2 obstruction introduced in our paper is distinct from all of these: it is a purely integral cohomological invariant of the G -lattice M , not a geometric invariant of the function field $F(M)^G$.

In this paper we give a complete answer:

<i>Property</i>	<i>Status</i>	<i>Method</i>
Rational	No (char 0)	$H^2(H, M _H) \cong C_9$; rational $\Rightarrow$ stably rational
Stably rational	No (char 0)	$H^2(H, M _H) \cong C_9$
Retract rational	Yes (any field)	Saltman [?, Corollary 3.13]

The main new ingredient is the computation of the second integral cohomology group of the restricted Procesi lattice $M|_H$: $H^2(H, M|_H) \cong C_9$ (Lemma 4.2). The exponent 9 of this group directly obstructs stable permutation: we prove that for any permutation $\mathbb{Z}[H]$ -lattice P , every element of $H^2(H, P)$ has order dividing 3 (Proposition 4.4). Consequently $M|_H$ cannot be stably permutation, and by Theorem 2.6, $Z_H(F, 9)$ is not stably rational over F (Theorem 4.7). This integral cohomological obstruction appears to be new for multiplicative invariant fields.

The logical chain is:

$$\begin{aligned}
& H^2(H, M|_H) \cong C_9 \\
& \implies M|_H \text{ is not stably permutation} && \text{(Proposition 4.5)} \\
& \implies Z_H(F, 9) \text{ is not stably rational} && \text{(Theorem 2.6)} \\
& \implies Z_H(F, 9) \text{ is not rational} && \text{(rationality} \Rightarrow \text{stable rationality)}.
\end{aligned}$$

Thus both non-stable-rationality and non-rationality hold over any field of characteristic 0 (via Theorem 2.6), without reliance on the open converse Endo–Miyata implication.

As complementary evidence, we compute $\text{Ext}_{\mathbb{Z}[H]}^1(I[H], M|_H) \cong C_9 \neq C_3 \times C_3 \cong \text{Ext}_{\mathbb{Z}[H]}^1(I[H], 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z})$ (Proposition 4.9), which provides an independent verification that $M|_H$ is not a permutation lattice.

Finally, we prove that $Z_H(F, 9)$ is retract rational over F (Theorem 5.3), via Saltman’s retract rationality theorem [?, Corollary 3.13].

We also compute the Bogomolov multiplier $B_0(C_3 \times C_3) \cong C_3 \neq 0$ (Proposition 4.12). The classical Bogomolov obstruction $B_0(H) \cong C_3$ rules out rationality of the Noether problem $F(H)^H$, but does not directly govern $Z_H(F, 9) = F(M|_H)^H$ (which is a different multiplicative invariant field). Thus the integral H^2 obstruction detects a phenomenon inaccessible to the classical Bogomolov–Saltman invariants (see §4.4).

Hypotheses. The original statement in [?, Problem 5.2] is made over an arbitrary field. The cohomological computations ($H^2(H, M|_H) \cong C_9$ and $\exp(H^2(H, P)) \mid 3$) are purely integral, valid whenever $\text{char } F \nmid 9$. The geometric conclusions (non-stable-rationality, non-rationality) require $\text{char } F = 0$ via Theorem 2.6, which depends on resolution of singularities.

Retract rationality holds over any field (Theorem 5.3). Neither non-stable-rationality nor non-rationality relies on the open converse Endo–Miyata implication.

Generalizability. The proof generalizes to all odd primes p : for $H = C_p \times C_p \subset S_{p^2}$ acting regularly, the identical cohomological argument gives $H^2(H, M|_H) \cong C_{p^2}$ while permutation lattices have exponent dividing p ; hence $Z_H(F, p^2)$ is not stably rational over F (Theorem 5.4). Appendices 6.2 and 6.3 provide computational verification for $p = 3$ and $p = 5$.

Notation. C_n denotes the cyclic group of order n . For the main case (Problem 5.2), $H = C_3 \times C_3 = \langle a, b \mid a^3 = b^3 = 1, ab = ba \rangle$; the generalization to $H = C_p \times C_p$ is treated in §5.1. The base field F is of characteristic 0. For a finite group G , a G -lattice is a finitely generated $\mathbb{Z}[G]$ -module that is $\mathbb{Z}$ -free. The group ring is denoted $\mathbb{Z}[G]$. The augmentation ideal is $I[G] = \ker(\mathbb{Z}[G] \rightarrow \mathbb{Z})$, where the augmentation map sends $\sum n_g g \mapsto \sum n_g$. Cohomology $H^*(G, M)$ denotes ordinary group cohomology; for $\mathbb{Z}[G]$ -lattices this coincides with Tate cohomology in positive degrees [?, Def. 6.2.4]. The notation $\exp(A)$ denotes the exponent of a finite abelian group A (the least positive integer annihilating A).

Declaration of Artificial Intelligence Assistance. In the preparation of this work, we used DeepSeek for initial brainstorming of the literature search strategy, drafting a preliminary version, and proofreading. All AI-generated content was critically reviewed, substantially rewritten, and fact-checked by the authors. The authors take full responsibility for scientific integrity of the entire manuscript.

Acknowledgements. We are supported by Xiangtan University project 09KZ-KZ08101 *algebraic invariants and local-global principles*.

2 Preliminaries

Definition 2.1. A $\mathbb{Z}[G]$ -lattice is a finitely generated $\mathbb{Z}[G]$ -module that is $\mathbb{Z}$ -free. A $\mathbb{Z}[G]$ -lattice M is called:

1. *permutation* if $M \cong \bigoplus_{i=1}^r \mathbb{Z}[G/K_i]$ for subgroups $K_i \leq G$, where $\mathbb{Z}[G/K]$ denotes the $\mathbb{Z}[G]$ -module with $\mathbb{Z}$ -basis the left cosets G/K and G -action by left multiplication;
2. *stably permutation* if there exist permutation lattices P, Q such that $M \oplus P \cong Q$;

These concepts are standard; see [?, §1].

Lemma 2.2 (Character of a permutation module). *Let $K \leq H$ be finite groups and let $\mathbb{Z}[H/K]$ be the permutation $\mathbb{Z}[H]$ -lattice (Definition 2.1). The character of the rational representation $\mathbb{Q}[H/K]$ is*

$$\chi_{H/K}(g) = |\{xK \in H/K : gxK = xK\}| = \begin{cases} [H : K], & g \in K, \\ 0, & g \notin K. \end{cases}$$

Moreover, for an abelian group H , the multiplicities m_K in a permutation lattice $P \cong \bigoplus_{K \leq H} m_K \cdot \mathbb{Z}[H/K]$ are uniquely determined by the character of $P \otimes_{\mathbb{Z}} \mathbb{Q}$ [?, Example 10.4].

This lemma is the key tool in Lemma 3.8 below, where the multiplicities of a permutation lattice with a given rational character are determined uniquely from character evaluations.

Definition 2.3. A $\mathbb{Z}[G]$ -lattice F is *flasque* if $\hat{H}^{-1}(K, F) = 0$ for all subgroups $K \leq G$ (equivalently, $\text{Ext}_G^1(F, P) = 0$ for every permutation lattice P) [?, §1, Lemme 9]. Here $\hat{H}^{-1}$ denotes Tate cohomology in degree -1 ; for a finite group G and a $\mathbb{Z}[G]$ -lattice M , $\hat{H}^{-1}(G, M) = \ker(N_G)/I[G]M$, where $N_G = \sum_{g \in G} g$ is the norm element and $I[G] = \ker(\mathbb{Z}[G] \rightarrow \mathbb{Z})$ is the augmentation ideal.

Definition 2.4. Two $\mathbb{Z}[G]$ -lattices M, N are called *similar* (written $M \sim N$) if $M \oplus S_1 \cong N \oplus S_2$ for some permutation lattices S_1, S_2 (Definition 2.1). This is an equivalence relation; denote the similarity class of M by $[M]$. Let

$$C(G)/S(G) = \{[M] : M \text{ a finitely generated } \mathbb{Z}[G]\text{-lattice}\},$$

a commutative monoid under direct sum, with zero element the class of permutation lattices. Its maximal subgroup is

$$D(G)/S(G) = \{[M] \in C(G)/S(G) : M \text{ is stably permutation}\},$$

which is the semigroup $C(\Pi)/S(\Pi)$ of [?, §4.7].

Proposition 2.5 (Flasque resolution). *Every $\mathbb{Z}[G]$ -lattice M has a flasque resolution*

$$0 \longrightarrow M \longrightarrow P \longrightarrow F \longrightarrow 0$$

with P permutation (Definition 2.1) and F flasque (Definition 2.3) [?, §1, Lemme 3]. The similarity class $[F] \in D(G)/S(G)$ (Definition 2.4) depends only on M [?, §1, Lemme 5]; we denote it by $\rho(M) = [F]$ and call it the flasque class of M . By construction, M is stably permutation iff $\rho(M) = 0$; this is the defining property of the group $D(G)/S(G)$.

Theorem 2.6 (Endo–Miyata, Colliot-Thélène–Sansuc, Voskresenskii). *Let G be a finite group and M a $\mathbb{Z}[G]$ -lattice on which G acts faithfully. Let F be a field of characteristic 0. Here $F(M)$ denotes the field of rational functions of the torus with character lattice M (the fraction field of the group algebra $F[M]$, a purely transcendental extension of F), on which G acts via the $\mathbb{Z}[G]$ -module structure of M ; $F(M)^G$ is its G -invariant subfield. Then $F(M)^G$ is stably rational over F if and only if M is a stably permutation $\mathbb{Z}[G]$ -lattice [?, 4.7, theorem 2] (i.e., its flasque class $\rho(M) \in D(G)/S(G)$ vanishes [?, §1, Lemme 6]).*

In the sequel we apply this theorem in the contrapositive direction: if $M|_H$ is *not* stably permutation, then $F(M|_H)^H \cong Z_H(F, 9)$ is *not* stably rational. The cohomological computation of §4 proves that $H^2(H, M|_H) \cong C_9$ while permutation lattices have H^2 -exponent dividing 3; hence $M|_H$ is not stably permutation (Theorem 4.6), and the theorem yields non-stable-rationality (Corollary 4.7). The same strategy generalizes to $C_p \times C_p$ in §5.1.

Theorem 2.7 (Universal Coefficient Theorem). *For a finite group G and a trivial G -module M , there is a natural short exact sequence*

$$0 \longrightarrow \mathrm{Ext}_{\mathbb{Z}}^1(H_{n-1}(G, \mathbb{Z}), M) \longrightarrow H^n(G, M) \longrightarrow \mathrm{Hom}(H_n(G, \mathbb{Z}), M) \longrightarrow 0$$

for all $n \geq 1$ [?, 3.6.5]. If M is divisible (e.g. $M = \mathbb{Q}/\mathbb{Z}$), the sequence splits, giving a direct sum decomposition.

This theorem is used in Proposition 4.12 to compute $H^2(H, \mathbb{Q}/\mathbb{Z})$.

3 The Procesi Lattice

We give a self-contained construction of the Procesi $\mathbb{Z}[S_9]$ -lattice M and the Procesi isomorphism, keeping only what is needed for the rationality computations. An accessible modern exposition is in [?, §5] and [?]; the original reference is [?].

3.1 Construction over S_9

Let $n = 9$ and let $\Omega = S_9/S_8$ be the set of left cosets of S_8 in S_9 . Then $|\Omega| = 9$ and S_9 acts on Ω by left multiplication. Define the permutation $\mathbb{Z}[S_9]$ -module

$$U = \mathbb{Z}[\Omega] \cong \bigoplus_{\omega \in \Omega} \mathbb{Z} \cdot e_{\omega},$$

where $\{e_\omega\}_{\omega \in \Omega}$ is the standard basis and $g \cdot e_\omega = e_{g\omega}$ for $g \in S_9$. Consider the $\mathbb{Z}[S_9]$ -module map

$$\varphi : U \otimes_{\mathbb{Z}} U \longrightarrow \mathbb{Z}[\Omega], \quad \varphi(e_\alpha \otimes e_\beta) = e_\alpha - e_\beta,$$

extended $\mathbb{Z}$ -linearly. The map φ is S_9 -equivariant:

$$\varphi(g \cdot (e_\alpha \otimes e_\beta)) = \varphi(e_{g\alpha} \otimes e_{g\beta}) = e_{g\alpha} - e_{g\beta} = g \cdot (e_\alpha - e_\beta) = g \cdot \varphi(e_\alpha \otimes e_\beta).$$

The image of φ is the *augmentation submodule*

$$I[\Omega] = \ker\left(\varepsilon : \mathbb{Z}[\Omega] \rightarrow \mathbb{Z}, \varepsilon\left(\sum a_\omega e_\omega\right) = \sum a_\omega\right).$$

Indeed, every difference $e_\alpha - e_\beta$ has augmentation zero, and conversely any element $\sum a_\omega e_\omega$ with $\sum a_\omega = 0$ is a $\mathbb{Z}$ -linear combination of such differences (take $\{e_\alpha - e_{\omega_0} : \alpha \neq \omega_0\}$ for a fixed ω_0). Moreover, φ is surjective onto $I[\Omega]$.

Definition 3.1. The *Procesi $\mathbb{Z}[S_9]$ -lattice* is $M = \ker(\varphi)$.

By construction we have the short exact sequence of $\mathbb{Z}[S_9]$ -modules

$$0 \longrightarrow M \longrightarrow U \otimes_{\mathbb{Z}} U \xrightarrow{\varphi} I[\Omega] \longrightarrow 0. \quad (1)$$

The module M is $\mathbb{Z}$ -free: it is a $\mathbb{Z}$ -submodule of the free $\mathbb{Z}$ -module $U \otimes_{\mathbb{Z}} U$, and every submodule of a free $\mathbb{Z}$ -module is free (since $\mathbb{Z}$ is a principal ideal domain) [?, Ch.VII, §3, Th.1, Cor.2]. The ranks are

$$\mathrm{rk}_{\mathbb{Z}} U = 9, \quad \mathrm{rk}_{\mathbb{Z}}(U \otimes U) = 81, \quad \mathrm{rk}_{\mathbb{Z}} I[\Omega] = 8, \quad \mathrm{rk}_{\mathbb{Z}} M = 73.$$

Remark 3.2. Procesi [?] proved that the center of the generic matrix algebra satisfies $Z(F, 9) \cong F(M)^{S_9}$, where $F(M)^{S_9}$ is the invariant field of the purely transcendental extension $F(M)$ under the natural S_9 -action induced by the $\mathbb{Z}[S_9]$ -module structure of M . For any subgroup $H \subseteq S_9$, taking H -invariants gives

$$Z_H(F, 9) = F(M)^H \cong F(M|_H)^H,$$

where $M|_H$ denotes M viewed as a $\mathbb{Z}[H]$ -module by restriction. This isomorphism holds over any field F of characteristic 0; see [?, §5], [?], and the references therein. For the computations that follow, only the exact sequence (1) and this identification are required.

3.2 Restriction to $H = C_3 \times C_3$

Now fix a transitive embedding $H = C_3 \times C_3 \hookrightarrow S_9$, i.e., an embedding whose image acts transitively on $\Omega = S_9/S_8$. Since $|H| = 9 = |\Omega|$, the action is moreover regular (i.e., free and transitive, called *simply transitive* in [?, ch.I, §5, no.6, def.7]): choosing a basepoint $\omega_0 \in \Omega$, the map $H \rightarrow \Omega$, $h \mapsto h \cdot \omega_0$, is an H -equivariant bijection. Consequently we obtain an explicit isomorphism of $\mathbb{Z}[H]$ -modules

$$U|_H \xrightarrow{\cong} \mathbb{Z}[H], \quad e_{h \cdot \omega_0} \mapsto h \quad (h \in H). \quad (2)$$

Proposition 3.3 (Independence of basepoint choice). *The isomorphism (2) is independent of the choice of ω_0 up to a $\mathbb{Z}[H]$ -module automorphism of $\mathbb{Z}[H]$. Consequently, $\text{Ext}_{\mathbb{Z}[H]}^1(-, M|_H)$, $H^2(H, M|_H)$, and the isomorphism class of $M|_H$ are independent of this choice.*

Proof. Let $\omega'_0 = g \cdot \omega_0$ be another basepoint. The isomorphism induced by ω'_0 sends $e_{h \cdot \omega'_0} = e_{hg \cdot \omega_0} \mapsto h$, while the original sends $e_{hg \cdot \omega_0} \mapsto hg$; hence the two isomorphisms differ by right multiplication by g^{-1} on $\mathbb{Z}[H]$. Right multiplication by a fixed element of H is a $\mathbb{Z}[H]$ -module automorphism of the left regular module. Therefore the induced map on cohomology and Ext^1 is the identity, and the isomorphism class of $M|_H = \ker(\pi|_H)$ is unchanged. $\square$

Proposition 3.4 (Independence of the embedding). *Let $\iota_1, \iota_2 : H \hookrightarrow S_9$ be two transitive embeddings. Then $M|_{\iota_1(H)} \cong M|_{\iota_2(H)}$ as $\mathbb{Z}[H]$ -modules.*

Proof. Both ι_1 and ι_2 make Ω into a regular H -set (since $|H| = 9 = |\Omega|$ and each action is transitive). Any two regular H -sets of the same cardinality are H -equivariantly isomorphic: fix basepoints $\omega_1, \omega_2 \in \Omega$, and define a bijection $\Omega \rightarrow \Omega$ by $h \cdot \omega_1 \mapsto h \cdot \omega_2$. This bijection is an element $\sigma \in S_9$, and by construction $\sigma \iota_1(h) \sigma^{-1} = \iota_2(h)$ for all $h \in H$. Conjugating the embedding by σ replaces the H -action on $U \otimes U$ with its conjugate under the S_9 -action, which induces an isomorphism $M|_{\iota_1(H)} \cong M|_{\iota_2(H)}$ of $\mathbb{Z}[H]$ -modules via the identification of H with $\iota_1(H)$ and $\iota_2(H)$ followed by the conjugation $\iota_2^{-1} \circ \iota_1$. $\square$

Proposition 3.5. *There is an isomorphism of $\mathbb{Z}[H]$ -modules*

$$\mathbb{Z}[H] \otimes_{\mathbb{Z}} \mathbb{Z}[H] \cong \mathbb{Z}[H]^{\oplus 9},$$

where the left side carries the diagonal action $h \cdot (x \otimes y) = hx \otimes hy$ and the right side carries componentwise left multiplication.

Proof. Identify $\mathbb{Z}[H]$ with the induced module $\text{Ind}_1^H(\mathbb{Z}) = \mathbb{Z}[H] \otimes_{\mathbb{Z}} \mathbb{Z}$, where Ind_1^H denotes induction from the trivial subgroup $\{1\}$ to H , and Res_1^H denotes restriction from H to $\{1\}$. For $\mathbb{Z}$ -modules A, B with trivial H -action, the projection formula for induced modules [?, Corollary 10.20] states

$$\text{Ind}_1^H(A) \otimes_{\mathbb{Z}} \text{Ind}_1^H(B) \cong \text{Ind}_1^H(A \otimes_{\mathbb{Z}} \text{Res}_1^H \text{Ind}_1^H(B))$$

as $\mathbb{Z}[H]$ -modules (diagonal action on the left, induced action on the right). Taking $A = B = \mathbb{Z}$ gives

$$\mathbb{Z}[H] \otimes_{\mathbb{Z}} \mathbb{Z}[H] \cong \text{Ind}_1^H(\text{Res}_1^H \text{Ind}_1^H(\mathbb{Z})).$$

The restriction $\text{Res}_1^H \text{Ind}_1^H(\mathbb{Z})$ is the regular module $\mathbb{Z}[H]$ viewed merely as a $\mathbb{Z}$ -module (forgetting the H -action); by definition $\mathbb{Z}[H]$ has $\mathbb{Z}$ -basis $\{h : h \in H\}$, so $\text{Res}_1^H \text{Ind}_1^H(\mathbb{Z}) \cong \mathbb{Z}^{\oplus 9}$. Applying Ind_1^H then yields $\text{Ind}_1^H(\mathbb{Z}^{\oplus 9}) \cong \text{Ind}_1^H(\mathbb{Z})^{\oplus 9} \cong \mathbb{Z}[H]^{\oplus 9}$. The isomorphism is H -equivariant: choose the $\mathbb{Z}$ -basis $\{g\}_{g \in H}$ of the second tensor factor $\mathbb{Z}[H]$; the map $x \otimes g \mapsto x$ in the g -th coordinate satisfies $h \cdot (x \otimes g) = hx \otimes hg \mapsto hx$ in the hg -th coordinate. $\square$

Under the identification (2) and Proposition 3.5, the restriction of the map φ to H yields the following.

Proposition 3.6 (Procesi sequence for H). *There is an exact sequence of $\mathbb{Z}[H]$ -modules*

$$0 \longrightarrow M|_H \longrightarrow \mathbb{Z}[H]^{\oplus 9} \xrightarrow{\pi|_H} I[H] \longrightarrow 0, \quad (3)$$

where $I[H] = \ker(\varepsilon : \mathbb{Z}[H] \rightarrow \mathbb{Z})$, $\varepsilon(\sum a_h h) = \sum a_h$, is the augmentation ideal of H . All modules in the sequence are finitely generated and $\mathbb{Z}$ -free, and $\text{rk}_{\mathbb{Z}} M|_H = 73$.

Proof. Exactness. Under the identifications, the map φ becomes a $\mathbb{Z}[H]$ -module map $\pi|_H : \mathbb{Z}[H]^{\oplus 9} \rightarrow I[H]$. Each standard basis element of $\mathbb{Z}[H]^{\oplus 9}$ maps to a difference $h - h'$ in $\mathbb{Z}[H]$; these differences generate the $\mathbb{Z}$ -module $I[H]$, and $I[H]$ is generated as a $\mathbb{Z}[H]$ -module by any single element $h - 1$ with $h \neq 1$; hence $\pi|_H$ is surjective. Define $M|_H = \ker(\pi|_H)$, the restriction of the Procesi lattice to H .

Freeness. All modules in the sequence are finitely generated and $\mathbb{Z}$ -free: $\mathbb{Z}[H]^{\oplus 9}$ is $\mathbb{Z}$ -free with basis the group elements in each coordinate; $I[H]$ is a $\mathbb{Z}$ -submodule of $\mathbb{Z}[H]$, hence free [?, Ch.VII, §3, Th.1, Cor.2]; $M|_H = \ker(\pi|_H)$ is a $\mathbb{Z}$ -submodule of $\mathbb{Z}[H]^{\oplus 9}$, hence free.

Rank. $\text{rk}_{\mathbb{Z}} M|_H = \text{rk}_{\mathbb{Z}} \mathbb{Z}[H]^{\oplus 9} - \text{rk}_{\mathbb{Z}} I[H] = 9 \cdot 9 - 8 = 73$, matching the rank of M over S_9 . $\square$

Proposition 3.7 (Rational character of $M|_H$).

$$M|_H \otimes_{\mathbb{Z}} \mathbb{Q} \cong 8 \cdot \mathbb{Q}[H] \oplus \mathbb{Q},$$

where $\mathbb{Q}$ denotes the one-dimensional trivial $\mathbb{Q}[H]$ -module.

Proof. Tensor the exact sequence of Proposition 3.6 with $\mathbb{Q}$ (flat over $\mathbb{Z}$, [?, Ch.II, §2, no.4, Th.1]):

$$0 \longrightarrow M|_H \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \mathbb{Q}[H]^{\oplus 9} \longrightarrow I[H] \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow 0.$$

The augmentation sequence

$$0 \longrightarrow I[H] \longrightarrow \mathbb{Z}[H] \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0 \quad (4)$$

remains exact after tensoring with $\mathbb{Q}$ (flat over $\mathbb{Z}$, [?, Ch.II, §2, no.4, Th.1]), giving

$$0 \longrightarrow I[H] \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \mathbb{Q}[H] \xrightarrow{\varepsilon} \mathbb{Q} \longrightarrow 0.$$

This sequence splits since $|H| = 9$ is invertible in $\mathbb{Q}$ (Maschke's Theorem [?, (3.14)]; the splitting is given by the idempotent $e_0 = \frac{1}{|H|} \sum_{h \in H} h$). Hence in the Grothendieck group $K_0(\mathbb{Q}[H])$ [?, (16.3)],

$$[I[H] \otimes_{\mathbb{Z}} \mathbb{Q}] = [\mathbb{Q}[H]] - [\mathbb{Q}].$$

From the exact sequence of $\mathbb{Q}[H]$ -modules above,

$$\begin{aligned} [M|_H \otimes \mathbb{Q}] &= 9[\mathbb{Q}[H]] - [I[H] \otimes \mathbb{Q}] \\ &= 9[\mathbb{Q}[H]] - ([\mathbb{Q}[H]] - [\mathbb{Q}]) \\ &= 8[\mathbb{Q}[H]] + [\mathbb{Q}]. \end{aligned}$$

Since $\mathbb{Q}[H]$ is semisimple by Maschke [?, (3.14)], equality in $K_0(\mathbb{Q}[H])$ implies

$$M|_H \otimes_{\mathbb{Z}} \mathbb{Q} \cong 8 \cdot \mathbb{Q}[H] \oplus \mathbb{Q}$$

of $\mathbb{Q}[H]$ -modules. □

From Proposition 3.7 we deduce: if $M|_H$ were a permutation $\mathbb{Z}[H]$ -lattice, its rational character would force $M|_H \cong 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}$.

Lemma 3.8. *Let P be a permutation $\mathbb{Z}[H]$ -lattice satisfying $P \otimes_{\mathbb{Z}} \mathbb{Q} \cong 8 \cdot \mathbb{Q}[H] \oplus \mathbb{Q}$. Then $P \cong 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}$.*

Proof. Since P is a permutation $\mathbb{Z}[H]$ -lattice (Definition 2.1), we can write

$$P \cong \bigoplus_{K \leq H} m_K \cdot \mathbb{Z}[H/K], \quad m_K \in \mathbb{Z}_{\geq 0},$$

where K runs over the subgroups of $H = C_3 \times C_3$. The subgroups are: $\{1\}$ (trivial, index 9); four subgroups of order 3 (each index 3), namely $K_1 = \langle a \rangle$, $K_2 = \langle b \rangle$, $K_3 = \langle ab \rangle$, $K_4 = \langle ab^2 \rangle$ where $H = \langle a \rangle \times \langle b \rangle \cong C_3 \times C_3$; and H itself (index 1, giving the trivial module $\mathbb{Z}$).

Character-theoretic determination of multiplicities. By Lemma 2.2, the character of $\mathbb{Q}[H/K]$ is given by

$$\chi_{H/K}(g) = \begin{cases} [H : K], & g \in K, \\ 0, & g \notin K, \end{cases}$$

and the multiplicities m_K are uniquely determined by the character of P .

By Proposition 3.7, $\chi_P = 8\chi_{H/\{1\}} + \chi_{H/H}$, so $\chi_P(1) = 8 \cdot 9 + 1 = 73$ and $\chi_P(g) = 0 + 1 = 1$ for all $g \neq 1$.

System of equations. Let $a = m_{\{1\}}$, $b_j = m_{K_j}$ ($j = 1, \dots, 4$), and $c = m_H$. Evaluating χ_P at the identity and at $g \in K_j \setminus \{1\}$ gives

$$\chi_P(1) = 9a + 3(b_1 + b_2 + b_3 + b_4) + c = 73, \quad (5)$$

$$\chi_P(g) = 3b_j + c = 1 \quad (g \in K_j, g \neq 1, j = 1, \dots, 4). \quad (6)$$

Solution. Equation (6) gives $b_j = (1-c)/3$. Since the multiplicities b_j are non-negative integers by definition of a permutation lattice decomposition, we must have $c = 1$, forcing $b_j = 0$ for all j . Substituting into (5): $9a + 0 + 1 = 73$, hence $a = 8$. Therefore $P \cong 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}$. $\square$

4 Cohomological Obstruction to Stable Permutation

In this section we compute the second integral cohomology group $H^2(H, M|_H)$ and prove that its exponent obstructs stable permutation.

Notation. Throughout this section we write $H^q(G, M)$ for the q -th integral group cohomology of a G -module M . The upright H is reserved for the cohomology functor, while the italic H denotes the group $H = C_3 \times C_3$.

4.1 Cohomology Computations

Lemma 4.1. *The group $H^1(H, I[H])$ is cyclic of order 9.*

Proof. Apply group cohomology $H^*(H, -)$ to the augmentation exact sequence (4). The associated long exact cohomology sequence reads

$$\begin{aligned} 0 \longrightarrow H^0(H, I[H]) \longrightarrow H^0(H, \mathbb{Z}[H]) \longrightarrow H^0(H, \mathbb{Z}) \\ \longrightarrow H^1(H, I[H]) \longrightarrow H^1(H, \mathbb{Z}[H]) \longrightarrow \dots \end{aligned}$$

We analyse the low-degree terms:

- $H^0(H, \mathbb{Z}[H]) = \mathbb{Z}[H]^H = \mathbb{Z} \cdot N_H$, where $N_H = \sum_{h \in H} h$ is the norm element.
- $H^0(H, \mathbb{Z}) = \mathbb{Z}^H = \mathbb{Z}$.
- The map $H^0(H, \mathbb{Z}[H]) \rightarrow H^0(H, \mathbb{Z})$ sends $N_H \mapsto \varepsilon(N_H) = \sum_{h \in H} 1 = |H| = 9$.
- $H^1(H, \mathbb{Z}[H]) = 0$: $\mathbb{Z}[H] \cong \text{Ind}_1^H \mathbb{Z}$ as $\mathbb{Z}[H]$ -modules, and Shapiro's Lemma [?, Theorem 6.3.2] gives $H^1(H, \text{Ind}_1^H \mathbb{Z}) \cong H^1(\{1\}, \mathbb{Z}) = 0$.

From the exactness at $H^0(H, \mathbb{Z})$ we obtain $\text{coker}(\mathbb{Z} \cdot N_H \xrightarrow{\varepsilon} \mathbb{Z}) = \mathbb{Z}/9\mathbb{Z}$, and exactness at $H^1(H, I[H])$ with the vanishing of $H^1(H, \mathbb{Z}[H])$ yields the short exact sequence

$$0 \longrightarrow \mathbb{Z}/9\mathbb{Z} \longrightarrow H^1(H, I[H]) \longrightarrow 0,$$

hence $H^1(H, I[H]) \cong \mathbb{Z}/9\mathbb{Z} \cong C_9$. □

Lemma 4.2. *For the Procesi restriction $M|_H$, $H^2(H, M|_H) \cong C_9$.*

Proof. Apply $H^*(H, -)$ to the exact sequence of Proposition 3.6:

$$0 \longrightarrow M|_H \longrightarrow \mathbb{Z}[H]^{\oplus 9} \longrightarrow I[H] \longrightarrow 0.$$

The associated long exact cohomology sequence gives

$$\begin{aligned} \dots \longrightarrow H^1(H, \mathbb{Z}[H]^{\oplus 9}) \longrightarrow H^1(H, I[H]) \\ \longrightarrow H^2(H, M|_H) \longrightarrow H^2(H, \mathbb{Z}[H]^{\oplus 9}) \longrightarrow \dots \end{aligned}$$

By Shapiro's Lemma [?, Theorem 6.3.2], $H^i(H, \mathbb{Z}[H]) \cong H^i(\{1\}, \mathbb{Z}) = 0$ for all $i > 0$. Hence $H^1(H, \mathbb{Z}[H]^{\oplus 9}) = H^2(H, \mathbb{Z}[H]^{\oplus 9}) = 0$, and the long exact sequence collapses to the isomorphism

$$H^2(H, M|_H) \cong H^1(H, I[H]).$$

By Lemma 4.1, $H^1(H, I[H]) \cong C_9$. □

Lemma 4.3. *With the trivial H -action on $\mathbb{Z}$, $H^2(H, \mathbb{Z}) \cong C_3 \times C_3$.*

Proof. Apply $H^*(H, -)$ to the short exact sequence of trivial H -modules

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0.$$

The associated long exact sequence reads

$$\begin{aligned} 0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \\ \longrightarrow H^1(H, \mathbb{Z}) \longrightarrow H^1(H, \mathbb{Q}) \longrightarrow H^1(H, \mathbb{Q}/\mathbb{Z}) \\ \longrightarrow H^2(H, \mathbb{Z}) \longrightarrow H^2(H, \mathbb{Q}) \longrightarrow \dots \end{aligned}$$

Since $\mathbb{Q}$ is uniquely divisible, multiplication by 3 is an automorphism of $\mathbb{Q}$; hence $H^i(H, \mathbb{Q}) = 0$ for all $i > 0$ [?, Proposition 1.6.2(c)]. Substituting $H^1(H, \mathbb{Q}) = H^2(H, \mathbb{Q}) = 0$ yields $H^2(H, \mathbb{Z}) \cong H^1(H, \mathbb{Q}/\mathbb{Z})$. Since $\mathbb{Q}/\mathbb{Z}$ carries the trivial H -action, $H^1(H, \mathbb{Q}/\mathbb{Z}) = \text{Hom}(H, \mathbb{Q}/\mathbb{Z})$ [?, Corollary 6.4.6]. Now $\text{Hom}(H, \mathbb{Q}/\mathbb{Z}) \cong C_3 \times C_3$ because $\text{Hom}(C_3, \mathbb{Q}/\mathbb{Z}) \cong C_3$ ($\mathbb{Q}/\mathbb{Z}$ has a unique subgroup of order 3) and $\text{Hom}(-, \mathbb{Q}/\mathbb{Z})$ preserves finite direct sums [?, Ch.II, §1, no.6, Prop.6, Cor.1]. Hence $H^2(H, \mathbb{Z}) \cong C_3 \times C_3$. $\square$

4.2 Exponent of H^2 for Permutation Lattices

Proposition 4.4. *Let P be a permutation $\mathbb{Z}[H]$ -lattice. Then every element of $H^2(H, P)$ has order dividing 3; equivalently, $\exp(H^2(H, P)) \mid 3$.*

Proof. Write $P \cong \bigoplus_i \mathbb{Z}[H/K_i]$ for subgroups $K_i \leq H$. Since $H^2(H, -)$ is additive and therefore preserves finite direct sums [?, Proposition 3.3.4], $H^2(H, P) \cong \bigoplus_i H^2(H, \mathbb{Z}[H/K_i])$. By Shapiro's Lemma [?, Theorem 6.3.2], $\mathbb{Z}[H/K_i] \cong \text{Ind}_{K_i}^H(\mathbb{Z})$, hence $H^2(H, \mathbb{Z}[H/K_i]) \cong H^2(K_i, \mathbb{Z})$, where $\mathbb{Z}$ carries the trivial K_i -action. For each subgroup $K \leq H$ we compute $H^2(K, \mathbb{Z})$:

- $K = \{1\}$: $H^2(\{1\}, \mathbb{Z}) = 0$.
- $K \cong C_3$ (four subgroups of order 3): $H^2(C_3, \mathbb{Z}) \cong \text{Hom}(C_3, \mathbb{Q}/\mathbb{Z}) \cong C_3$.
- $K = H = C_3 \times C_3$: $H^2(H, \mathbb{Z}) \cong C_3 \times C_3$ (Lemma 4.3).

In every case $\exp(H^2(K, \mathbb{Z})) \mid 3$. Hence $\exp(H^2(H, P)) \mid 3$. $\square$

Proposition 4.5. *If a $\mathbb{Z}[H]$ -lattice L is stably permutation, then*

$$\exp(H^2(H, L)) \mid 3.$$

Proof. Write $L \oplus P \cong Q$ with P, Q permutation. By additivity [?, Proposition 3.3.4], $H^2(H, L) \oplus H^2(H, P) \cong H^2(H, Q)$. Both $H^2(H, P)$ and $H^2(H, Q)$ have exponent dividing 3 by Proposition 4.4. For any $x \in H^2(H, L)$, the element $(x, 0) \in H^2(H, Q)$ satisfies $3(x, 0) = 0$; hence $3x = 0$. Therefore $\exp(H^2(H, L)) \mid 3$. $\square$

4.3 The Obstruction

Theorem 4.6 (Non-stably-permutation). *The restricted Procesi lattice $M|_H$ is not a stably permutation $\mathbb{Z}[H]$ -lattice.*

Proof. By Lemma 4.2, $H^2(H, M|_H) \cong C_9$, a group of exponent 9. If $M|_H$ were stably permutation, Proposition 4.5 would force $\exp(H^2(H, M|_H)) \mid 3$. Contradiction. $\square$

Corollary 4.7 (Non-stably-rational). *The field $Z_H(F, 9)$ is not stably rational over F .*

Proof. Recall that $Z_H(F, 9) \cong F(M|_H)^H$ (Remark 3.2). By Theorem 4.6, $M|_H$ is not stably permutation. By Theorem 2.6, $F(M|_H)^H$ is stably rational over F if and only if $M|_H$ is stably permutation. Hence $Z_H(F, 9)$ is not stably rational. $\square$

Remark 4.8. The purely lattice-theoretic part (Theorem 4.6) holds integrally, hence for any field whose characteristic does not divide $|H| = 9$. The geometric conclusion (Theorem 4.7) additionally requires characteristic 0 via Theorem 2.6.

Proposition 4.9. *The restricted Procesi lattice $M|_H$ is not a permutation $\mathbb{Z}[H]$ -lattice.*

Proof. Apply $\text{Hom}_{\mathbb{Z}[H]}(-, L)$ to (4). The associated long exact sequence reads

$$\begin{aligned} 0 \longrightarrow \text{Hom}_{\mathbb{Z}[H]}(\mathbb{Z}, L) &\longrightarrow \text{Hom}_{\mathbb{Z}[H]}(\mathbb{Z}[H], L) \longrightarrow \text{Hom}_{\mathbb{Z}[H]}(I[H], L) \\ &\longrightarrow \text{Ext}_{\mathbb{Z}[H]}^1(\mathbb{Z}, L) \longrightarrow \text{Ext}_{\mathbb{Z}[H]}^1(\mathbb{Z}[H], L) \longrightarrow \text{Ext}_{\mathbb{Z}[H]}^1(I[H], L) \\ &\longrightarrow \text{Ext}_{\mathbb{Z}[H]}^2(\mathbb{Z}, L) \longrightarrow \text{Ext}_{\mathbb{Z}[H]}^2(\mathbb{Z}[H], L) \longrightarrow \cdots \end{aligned}$$

Since $\mathbb{Z}[H]$ is projective, $\text{Ext}_{\mathbb{Z}[H]}^i(\mathbb{Z}[H], -) = 0$ for all $i \geq 1$. Substituting this collapses the sequence to an isomorphism $\text{Ext}_{\mathbb{Z}[H]}^1(I[H], L) \cong \text{Ext}_{\mathbb{Z}[H]}^2(\mathbb{Z}, L) = H^2(H, L)$. Taking $L = M|_H$ and $L = 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}$ yields

$$\text{Ext}_{\mathbb{Z}[H]}^1(I[H], M|_H) \cong C_9, \quad \text{Ext}_{\mathbb{Z}[H]}^1(I[H], 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}) \cong C_3 \times C_3.$$

If $M|_H$ were isomorphic to $8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}$, these groups would coincide, contradicting $C_9 \not\cong C_3 \times C_3$. $\square$

Remark 4.10. Proposition 4.9 is strictly weaker than Theorem 4.6 (which proves non-stable permutation). The gap is significant: if M is a permutation lattice, then $F(M)^G$ is stably rational [?, Theorem 1.2]. The converse

“ $F(M)^G$ rational $\Rightarrow M$ stably permutation” is known for groups with cyclic Sylow subgroups [?, Theorem 1.5], but is *open* for groups with non-cyclic Sylow subgroups (including $H = C_3 \times C_3$). Therefore ruling out $M|_H$ being a permutation lattice does *not* imply non-rationality; that would require the open converse, whereas our main proof uses only the proved equivalence “stably rational $\Leftrightarrow$ stably permutation” (Theorem 2.6).

4.4 Comparison with Classical Birational Obstructions

We now situate the integral H^2 obstruction of §4 among the classical birational invariants. We compute the Bogomolov multiplier $B_0(C_3 \times C_3)$ and compare the H^2 -obstruction with five existing approaches:

1. the unramified Brauer group and Bogomolov multiplier;
2. the Hoshi–Kang–Yamasaki tables for multiplicative invariant fields of small dimension;
3. Swan–Endo–Miyata quasi-permutation modules;
4. Lenstra–Saltman Brauer-group invariants for the Noether problem;
5. the flasque class $\rho(M|_H)$ in $D(H)/S(H)$.

We also formulate necessary conditions for the H^2 -exponent obstruction and close with scope, limitations, and open directions.

Comparison with the Bogomolov multiplier.

Definition 4.11 (Bogomolov multiplier). For a finite group G ,

$$B_0(G) = \ker \left(H^2(G, \mathbb{Q}/\mathbb{Z}) \longrightarrow \bigoplus_{A < G} H^2(A, \mathbb{Q}/\mathbb{Z}) \right),$$

where the product runs over all *abelian* subgroups $A < G$, G acts trivially on $\mathbb{Q}/\mathbb{Z}$, and the map is the product of restriction homomorphisms [?, §4, Definition 1].

Proposition 4.12. $B_0(C_3 \times C_3) \cong C_3$.

Proof. Apply the Universal Coefficient Theorem 2.7 with $M = \mathbb{Q}/\mathbb{Z}$ and $n = 2$. Since $\mathbb{Q}/\mathbb{Z}$ is divisible, the sequence splits, giving

$$H^2(H, \mathbb{Q}/\mathbb{Z}) \cong \text{Hom}(H_2(H, \mathbb{Z}), \mathbb{Q}/\mathbb{Z}) \oplus \text{Ext}_{\mathbb{Z}}^1(H_1(H, \mathbb{Z}), \mathbb{Q}/\mathbb{Z}).$$

For $H = C_3 \times C_3$: $H_1(H, \mathbb{Z}) = H^{\text{ab}} = H \cong C_3 \times C_3$. By [?, Ch. V, Theorem 6.4(iii)], $H_2(H, \mathbb{Z}) \cong \bigwedge_{\mathbb{Z}}^2 H \cong C_3$. Hence

$$H^2(H, \mathbb{Q}/\mathbb{Z}) \cong \text{Hom}(C_3, \mathbb{Q}/\mathbb{Z}) \oplus (\text{Ext}(C_3, \mathbb{Q}/\mathbb{Z}) \oplus \text{Ext}(C_3, \mathbb{Q}/\mathbb{Z})) \cong (C_3)^3.$$

Identify $\text{Ext}(C_3 \times C_3, \mathbb{Q}/\mathbb{Z}) \cong \text{Hom}(C_3, \mathbb{Q}/\mathbb{Z})^{\oplus 2}$ via the projections $\pi_1, \pi_2 : C_3 \times C_3 \rightarrow C_3$: an element is a pair (φ_1, φ_2) with $\varphi_i \in \text{Hom}(C_3, \mathbb{Q}/\mathbb{Z}) \cong C_3$.

For a proper subgroup $K \cong C_3$ we compute analogously.

1. $H_1(K, \mathbb{Z}) \cong K^{\text{ab}} = K \cong C_3$ because for any group G , the first integral homology is the abelianization: $H_1(G, \mathbb{Z}) \cong G/[G, G]$ [?, Lemma 6.1.2]; since K is abelian, $K^{\text{ab}} = K$.
2. $H_2(K, \mathbb{Z}) = 0$ because for a finite cyclic group, the exterior algebra $\bigwedge_{\mathbb{Z}}^* K$ vanishes in degrees ≥ 2 ; hence by [?, Ch. V, Theorem 6.4(iii)], $H_2(K, \mathbb{Z}) \cong \bigwedge_{\mathbb{Z}}^2 K = 0$.
3. Applying the Universal Coefficient Theorem 2.7 (which splits because $\mathbb{Q}/\mathbb{Z}$ is divisible), the Hom term vanishes by step (2), yielding

$$H^2(K, \mathbb{Q}/\mathbb{Z}) \cong \text{Ext}_{\mathbb{Z}}^1(H_1(K, \mathbb{Z}), \mathbb{Q}/\mathbb{Z}) \cong \text{Ext}_{\mathbb{Z}}^1(C_3, \mathbb{Q}/\mathbb{Z}).$$

4. $\text{Ext}_{\mathbb{Z}}^1(C_3, \mathbb{Q}/\mathbb{Z}) \cong C_3$: the standard free resolution $0 \rightarrow \mathbb{Z} \xrightarrow{3} \mathbb{Z} \rightarrow C_3 \rightarrow 0$ gives $\text{Ext}_{\mathbb{Z}}^1(C_3, A) \cong A/3A$ for any abelian group A [?, Application 3.3.3]. Taking $A = \mathbb{Q}/\mathbb{Z}$, we have $(\mathbb{Q}/\mathbb{Z})/3(\mathbb{Q}/\mathbb{Z}) = (\mathbb{Q}/\mathbb{Z})[3]$, the 3-torsion subgroup, which is isomorphic to C_3 because $\mathbb{Q}/\mathbb{Z}$ has a unique subgroup of order 3 [?, Ch.II, §1, no.6, Prop.6, Cor.1].

Thus $H^2(K, \mathbb{Q}/\mathbb{Z}) \cong C_3$.

The restriction $\text{Res}_K^H : H^2(H, \mathbb{Q}/\mathbb{Z}) \rightarrow H^2(K, \mathbb{Q}/\mathbb{Z})$ is computed on the Ext part as follows. By functoriality of the abelianization isomorphism [?, Lemma 6.1.2], the inclusion $K \hookrightarrow H$ induces on H_1 the inclusion $K \hookrightarrow H$. Writing $K = \langle a^i b^j \rangle$, this is the map $\mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$ sending $1 \mapsto (i \bmod 3, j \bmod 3)$. On $\text{Ext}(-, \mathbb{Q}/\mathbb{Z})$, this sends (φ_1, φ_2) to $i\varphi_1 + j\varphi_2$ [?, Ch. III, §1]. For the four subgroups $K_1 = \langle a \rangle$, $K_2 = \langle b \rangle$, $K_3 = \langle ab \rangle$, $K_4 = \langle ab^2 \rangle$, we obtain

$$(\varphi_1, \varphi_2) \mapsto (\varphi_1, \varphi_2, \varphi_1 + \varphi_2, \varphi_1 + 2\varphi_2) \in (C_3)^{\oplus 4}.$$

This linear map $(\mathbb{F}_3)^2 \rightarrow (\mathbb{F}_3)^4$ has matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$, whose columns are linearly independent over $\mathbb{F}_3$; hence the map is injective. The Hom part vanishes under restriction to every proper K because $H_2(K, \mathbb{Z}) = 0$ by step (2) above.

The Bogomolov multiplier is the common kernel of all restrictions to proper subgroups:

$$B_0(H) = \ker\left(H^2(H, \mathbb{Q}/\mathbb{Z}) \rightarrow \bigoplus_{K < H} H^2(K, \mathbb{Q}/\mathbb{Z})\right).$$

The Ext part injects into the product over the four subgroups K_i ; the Hom part lies in the kernel because $H_2(K) = 0$ for all proper K . Therefore $B_0(H) = \text{Hom}(H_2(H, \mathbb{Z}), \mathbb{Q}/\mathbb{Z}) \cong C_3$. (This result is also known from the general formula $B_0(C_p \times C_p) \cong C_p$ for odd primes p ; see [?, §6, Lemma 1].) $\square$

Remark 4.13. $B_0(C_3 \times C_3) \neq 0$ implies that the Noether problem $F(H)^H$ has non-trivial unramified Brauer group (Bogomolov’s theorem [?, §4, Theorem 6]), hence is not retract rational. In contrast, $Z_H(F, 9) = F(M|_H)^H$ is retract rational (Theorem 5.3) but not stably rational (Theorem 4.7). The unramified Brauer group $\text{Br}_{\text{nr}}(Z_H(F, 9))/\text{Br}(F)$ is the portion of the flasque obstruction captured by a $\mathbb{Q}/\mathbb{Z}$ -valued invariant (cf. §5.1); we have shown that it is either C_3 or C_9 — non-trivial in any case — while the integral H^2 obstruction has exponent 9 regardless. Thus the integral obstruction is strictly finer than the Br_{nr} -obstruction: even if $\text{Br}_{\text{nr}} \cong \text{Br}(F) \times C_3$, the exponent 9 of $H^2(H, M|_H)$ still certifies non-stable-permutation.

Comparison with Hoshi–Kang–Yamasaki. Hoshi, Kang, and Yamasaki [?] have systematically computed unramified Brauer groups for multiplicative invariant fields of small groups; their tables cover groups with ≤ 6 variables. For groups of order 9, their results show that all unramified Brauer obstructions originate from $B_0(H) \cong C_3$. Our field $Z_H(F, 9)$ has transcendence degree 73 and is therefore outside the range of their classification. The integral H^2 obstruction C_9 is of a different nature: it operates at the level of $\mathbb{Z}[H]$ -lattices, whereas the HKY classification is based on Br_{nr} -valued invariants. Theorem 4.7 thus exhibits a multiplicative invariant field whose non-stable-rationality is detected by an integral cohomological obstruction, which appears to be new in this context.

Swan–Endo–Miyata modules. Swan [?, Theorem A] introduced the concept of a projective module over a group ring being *induced from a subgroup*; Endo–Miyata [?, §1] developed the theory of quasi-permutation modules. For a $\mathbb{Z}[H]$ -lattice L , the property “ L is a permutation lattice” is equivalent to “ $L \otimes_{\mathbb{Z}} \mathbb{Q}$ has a suitable character and L is $\mathbb{Z}[H]$ -projective” (by Rim’s theorem for p -groups). Our Ext^1 computation (Proposition 4.9) shows that $M|_H$ fails even the permutation-lattice test, but the main result is stronger: $M|_H$ is not *stably* permutation — a property that is not captured by Swan–Endo–Miyata invariants alone.

Lenstra–Saltman invariants. Lenstra [?] classified Noether problems via class groups of maximal orders and cyclic-by-cyclic extensions; Saltman [?, Theorem 3.2] used an invariant in $\text{Br}(F)/\text{Br}_0(F)$ to construct the first non-rational stably rational fields. These invariants operate at the level of the function field $F(M)^G$ and are valued in Brauer groups or class groups. By contrast, our obstruction operates at the level of the $\mathbb{Z}[H]$ -lattice M , before passing to the field of fractions. For $H = C_3 \times C_3$, Saltman’s Corollary 3.13 gives retract rationality unconditionally; the integral H^2 obstruction then shows that retract rationality is optimal: neither rationality nor stable rationality holds.

Flasque-class obstructions (CTS). Let $\rho(M|_H) \in D(H)/S(H)$ be the flasque invariant of $M|_H$ [?, §1]. By Theorem 2.6,

$$\rho(M|_H) \neq 0 \iff \text{not stably rational.}$$

Our exponent argument gives a *sufficient condition* for $\rho(M|_H) \neq 0$, computable from the Procesi sequence without an explicit flasque resolution. Specifically, if

$$\exp(H^2(H, M|_H)) > \max_{P \text{ perm}} \exp(H^2(H, P)),$$

then $M|_H$ is not stably permutation and therefore $\rho(M|_H) \neq 0$. No Swan subgroup, maximal order, or p -adic genus computation is needed.

Necessary conditions for the obstruction. The integral H^2 -exponent obstruction applies when a $\mathbb{Z}[G]$ -lattice M satisfies

$$\exp(H^2(G, M)) > \max_{K \leq G} \exp(H^2(K, \mathbb{Z})).$$

For an arbitrary finite group G , write every permutation lattice as $P = \bigoplus_i \mathbb{Z}[G/K_i]$; then

$$\exp(H^2(G, P)) \mid \max_{K \leq G} \exp(H^2(K, \mathbb{Z})).$$

Consequently any M whose H^2 -exponent exceeds this maximum is not stably permutation. Whether such M arise naturally for other multiplicative invariant fields (e.g. from non-regular embeddings or non-abelian G) is an open direction; see §5.1.

5 Retract Rationality and Conclusion

Proposition 5.1 (Faithfulness of the H -action). *Let $H = C_3 \times C_3$ act on $M_9(F)$ via the regular representation $H \hookrightarrow S_9$ by permutation of matrix*

indices: $h \cdot e_{ij} = e_{h(i),h(j)}$ for the standard matrix units. The action of H on $M|_H$ is the restriction of the diagonal left-regular action on $\mathbb{Z}[H]^{\oplus 9}$:

$$h \cdot (x_1, \dots, x_9) = (hx_1, \dots, hx_9), \quad h \in H, (x_1, \dots, x_9) \in M|_H \subset \mathbb{Z}[H]^{\oplus 9}.$$

This action is faithful.

Proof. The action on $M_9(F)$ is faithful because $h \neq 1$ implies $h(i) \neq i$ for some index i , whence $h \cdot e_{ij} \neq e_{ij}$. The Procesi isomorphism $Z_H(F, 9) \cong F(M|_H)^H$ (Remark 3.2) intertwines the H -actions. Any $h \in H$ acting trivially on $M|_H$ would, by functoriality of the field of fractions, act trivially on $F(M|_H) = \text{Frac}(\text{Sym}_F(M|_H \otimes_{\mathbb{Z}} F))$ and therefore on $F(M|_H)^H \cong Z_H(F, 9)$, hence on $M_9(F)$ — contradicting faithfulness on $M_9(F)$. $\square$

Theorem 5.2 (Saltman, 1984). [*?*, Corollary 3.13]. *Let A be a finite abelian group, and let 2^r be the highest power of 2 dividing the exponent of A . Assume F is a field such that either F has characteristic 2 or $F(\rho)/F$ is a cyclic extension, where ρ is a primitive 2^r -th root of unity. Let V be a finitely generated $F[A]$ -module on which A acts faithfully. Then $F(V)^A$ is retract rational over F .*

Corollary 5.3 (Retract rationality). *The field $Z_H(F, 9)$ is retract rational over F (Remark 3.2).*

Proof. $Z_H(F, 9) \cong F(M|_H)^H$ (Remark 3.2). Let $V = M|_H \otimes_{\mathbb{Z}} F$; then $F(V)^H \cong F(M|_H)^H$. Apply Theorem 5.2 with $A = H = C_3 \times C_3$. The exponent of H is 3 (odd), so $2^r = 1$ and $\rho = 1$; the condition on $F(1)/F$ is vacuous. The group H acts faithfully on V because it acts faithfully on $M|_H$ (Proposition 5.1) and $\text{char } F = 0$. Hence $Z_H(F, 9)$ is retract rational over F . $\square$

5.1 Generalization to $C_p \times C_p$

Theorem 5.4 (Obstruction for $C_p \times C_p$). *Let p be an odd prime and let $H = C_p \times C_p \subset S_{p^2}$ act regularly. Let $M|_H$ be the restriction of the Procesi $\mathbb{Z}[S_{p^2}]$ -lattice to H , defined by the exact sequence*

$$0 \longrightarrow M|_H \longrightarrow \mathbb{Z}[H]^{\oplus p^2} \longrightarrow I[H] \longrightarrow 0.$$

Let F be a field of characteristic 0, and let $Z_H(F, p^2)$ denote the center of the generic division algebra of degree p^2 over F (cf. Remark 3.2), with Procesi isomorphism $Z_H(F, p^2) \cong F(M|_H)^H$. Then:

1. $H^2(H, M|_H) \cong C_{p^2}$.

2. For any permutation $\mathbb{Z}[H]$ -lattice P , $\exp(H^2(H, P)) \mid p$.
3. Consequently, $M|_H$ is not stably permutation, and $Z_H(F, p^2)$ is not stably rational (hence not rational) over F .
4. $Z_H(F, p^2)$ is retract rational over F .

Proof. The Procesi restriction to H follows the same construction as in §3: the regular identification $U|_H \cong \mathbb{Z}[H]$ (Proposition 3.3) and the isomorphism $\mathbb{Z}[H] \otimes \mathbb{Z}[H] \cong \mathbb{Z}[H]^{\oplus p^2}$ (Proposition 3.5, whose proof works verbatim with 9 replaced by p^2) hold integrally for all p . The map $\pi|_H : \mathbb{Z}[H]^{\oplus p^2} \rightarrow I[H]$ is surjective because $I[H]$ is generated by $h - 1$ ($h \neq 1$), and $M|_H = \ker(\pi|_H)$ is $\mathbb{Z}$ -free (submodule of a free $\mathbb{Z}$ -module) of $\mathbb{Z}$ -rank $p^4 - p^2 + 1$.

(1) Apply $H^*(H, -)$ to the Procesi sequence. Shapiro's lemma and additivity [?, Proposition 3.3.4] give $H^i(H, \mathbb{Z}[H]^{\oplus p^2}) = 0$ for $i = 1, 2$. The long exact sequence therefore collapses to

$$H^2(H, M|_H) \cong H^1(H, I[H]).$$

From (4) we obtain

$$H^1(H, I[H]) \cong \operatorname{coker}(H^0(\mathbb{Z}[H]) \rightarrow H^0(\mathbb{Z})) = \mathbb{Z}/p^2\mathbb{Z} \cong C_{p^2}.$$

(2) Write $P \cong \bigoplus_i \mathbb{Z}[H/K_i]$. By Shapiro, $H^2(H, P) \cong \bigoplus_i H^2(K_i, \mathbb{Z})$. Subgroups of $C_p \times C_p$ are $\{1\}$ ($H^2 = 0$), C_p ($H^2 \cong C_p$), and $C_p \times C_p$ ($H^2 \cong C_p \times C_p$). Hence $\exp(H^2(H, P)) \mid p$.

(3) $\exp(H^2(M|_H)) = p^2 \nmid p \geq \exp(H^2(H, P))$ for any permutation P , so $M|_H$ is not stably permutation (otherwise stably permutation would force exponent dividing p , contradicting p^2). By Theorem 2.6, $Z_H(F, p^2) \cong F(M|_H)^H$ is not stably rational over F . Rational $\Rightarrow$ stably rational, so it is not rational either.

(4) Saltman [?, Corollary 3.13] applies verbatim: $H = C_p \times C_p$ is abelian of odd exponent, so the root-of-unity hypothesis is vacuous. The H -action on $M|_H \otimes_{\mathbb{Z}} F$ is faithful (Proposition 5.1). $\square$

Remark 5.5. The proof of Theorem 5.4 holds for all odd primes p without any dependence on $p = 3$ specifically: the regular embedding $H \hookrightarrow S_{p^2}$ (independent of the choice up to conjugation; Proposition 3.4), Proposition 3.5, the cohomological vanishing (Shapiro), and the subgroup analysis of $C_p \times C_p$ are all purely formal and independent of p . Appendices 6.2 and 6.3 provide computational verification for $p = 3$ and $p = 5$ via direct integer Smith normal form on the bar resolution cochain matrices. The converse Endo–Miyata implication (rational $\Rightarrow$ stably permutation for non-cyclic p -groups) remains open but is not required for any of the above conclusions.

We summarize our results on [?, Problem 5.2]:

<i>Property</i>	<i>Status</i>	<i>Method</i>
Rational	No (char 0)	$H^2(H, M _H) \cong C_9$; rational $\Rightarrow$ stably rational
Stably rational	No (char 0)	$H^2(H, M _H) \cong C_9$
Retract rational	Yes (any field)	Saltman [?, Corollary 3.13]

The central new result is $H^2(H, M|_H) \cong C_9$ (Lemma 4.2) and the exponent argument showing that this obstructs stable permutation (Theorem 4.6). The chain of implications

$$\begin{aligned}
& H^2 \cong C_9 \\
& \implies \text{not stably permutation} \quad (\exp \mid 3 \text{ for permutation lattices}) \\
& \implies \text{not stably rational} \quad (\text{Theorem 2.6}) \\
& \implies \text{not rational} \quad (\text{rational} \Rightarrow \text{stably rational})
\end{aligned}$$

resolves both the stable rationality and rationality questions simultaneously (in characteristic 0, without reliance on the open converse Endo–Miyata implication). The Ext^1 computation (Proposition 4.9) provides independent confirmation that $M|_H$ is not a permutation lattice. The Bogomolov multiplier computation $B_0(C_3 \times C_3) \cong C_3$ (Proposition 4.12) shows that the classical unramified Brauer group of the Noether problem $F(H)^H$ is non-zero, while the integral H^2 obstruction operates at the level of $\mathbb{Z}[H]$ -lattices and is of a different type.

Scope, limitations, and further directions. Generalization to $C_p \times C_p$.

The entire proof generalizes to all odd primes p (the cohomological part is purely integral; the geometric conclusion requires $\text{char } F = 0$) (see §6.3 above). For $p = 5$, a direct computational verification (Appendix 6.3) confirms the pattern: $H^1(H, \mathbb{Z}[H]) = 0$ (free rank 0, $\text{rank}(d^0) = 24$, $\text{rank}(d^1) = 601$), $H^1(H, I[H]) \cong C_{25}$ (free rank 0, $\text{rank}(d^0) = 24$, $\text{rank}(d^1) = 576$), $H^2(H, \mathbb{Z}) \cong C_5 \times C_5$, and consequently $H^2(H, M|_H) \cong C_{25}$ with exponent $25 \nmid 5$. All steps are purely formal and independent of p ; no class-group or p -adic integrality conditions intervene. The converse Endo–Miyata implication remains open for all non-cyclic p -groups, but is not required for the main results (since $\text{not stably perm} \implies \text{not stably rational} \implies \text{not rational}$).

Beyond regular embeddings. Our proof uses the regular embedding $H \hookrightarrow S_{p^2}$ (Proposition 3.4) in three essential ways. We itemize them to clarify what would change for other embeddings.

1. *Identification* $U|_H \cong \mathbb{Z}[H]$. This relies on the action of H on $\Omega = S_{p^2}/S_{p^2-1}$ being regular (free and transitive, hence H -equivariantly isomorphic to the left regular action), and is independent of the base-point up to a $\mathbb{Z}[H]$ -module automorphism (Proposition 3.3). If H acts on Ω with multiple orbits or with non-trivial stabilizers, then $U|_H \cong \bigoplus_{K \leq H} m_K \cdot \mathbb{Z}[H/K]$ where the multiplicities m_K depend on the orbit structure. Shapiro's lemma would then give $H^i(H, U|_H \otimes U|_H) \cong \bigoplus_{K, L} m_K m_L H^i(K \cap L, \mathbb{Z})$, generally non-zero for $i > 0$, so the connecting-homomorphism isomorphism $H^2 \cong H^1(I[H])$ would no longer be an isomorphism.
2. *Projection formula* $\mathbb{Z}[H] \otimes \mathbb{Z}[H] \cong \mathbb{Z}[H]^{\oplus p^2}$. This holds for the regular $\mathbb{Z}[H]$ -module and relies on $\text{Res}_1^H \text{Ind}_1^H \mathbb{Z} \cong \mathbb{Z}^{\oplus p^2}$. For a non-regular permutation module $\mathbb{Z}[H/K]$, the tensor product $\mathbb{Z}[H/K] \otimes \mathbb{Z}[H/L]$ decomposes via Mackey's formula into summands indexed by double cosets; the ranks and cohomological properties change.
3. *Surjectivity* $\mathbb{Z}[H]^{\oplus p^2} \rightarrow I[H]$. For the regular action, $I[H]$ is generated as a $\mathbb{Z}[H]$ -module by a single element $h - 1$, and the map from p^2 copies of $\mathbb{Z}[H]$ is surjective. For a non-regular permutation lattice, the image of the Procesi map φ is a submodule of $I[\Omega]$ that may be strictly smaller, requiring analysis of the specific orbit decomposition.

In summary, for a non-regular embedding the short exact sequence

$$0 \longrightarrow M|_H \longrightarrow P_1 \longrightarrow I[\Omega] \longrightarrow 0$$

(with P_1 a permutation lattice not necessarily of the form $\mathbb{Z}[H]^{\oplus n}$) remains exact, but the cohomological analysis of P_1 becomes substantially more involved and the simple exponent obstruction may no longer yield a definitive answer. Computing $H^2(H, M|_H)$ for one or two non-regular embeddings of $C_3 \times C_3$ into S_n (e.g. the intransitive embedding $C_3 \times C_3 \subset S_3 \times S_3 \subset S_6$) would be a natural next step.

Non-abelian p -groups. For non-abelian groups the situation is more complex. The subgroup cohomology classification used in Proposition 4.4 extends: if G is any finite group and $P = \bigoplus_i \mathbb{Z}[G/K_i]$ is a permutation lattice, then $\exp(H^2(G, P)) \mid \max_{K \leq G} \exp(H^2(K, \mathbb{Z}))$ by Shapiro's lemma. The obstruction therefore applies whenever one can find a natural G -lattice M with $\exp(H^2(G, M)) > \max_{K \leq G} \exp(H^2(K, \mathbb{Z}))$. For the Heisenberg group of order p^3 , nontrivial $H^2(G, \mathbb{Z}) \cong C_p \times C_p$ (by the five-term exact sequence and $H_2(G, \mathbb{Z}) \cong C_p \times C_p$ for extraspecial p -groups of exponent p), so the maximal

permutation exponent is p . Whether a natural M with $\exp(H^2(G, M)) = p^2$ exists for such groups (e.g. from a Procesi-like construction or from a multiplicative invariant field) is an open question.

Flasque class and unramified Brauer group. Let

$$0 \longrightarrow M|_H \longrightarrow P \longrightarrow F \longrightarrow 0$$

be a flasque resolution of $M|_H$ [?, §1, Lemme 3] (such a resolution always exists with P permutation and F flasque). The flasque class is $\rho(M|_H) = [F] \in D(H)/S(H)$; since $M|_H$ is not stably permutation, $\rho(M|_H) \neq 0$.

Flasqueness vs. H^1 . For non-cyclic $H = C_3 \times C_3$, the flasque condition $\hat{H}^{-1}(K, F) = 0$ for all $K \leq H$ [?, §1, Lemme 1(iii)] is *not* equivalent to $H^1(K, F) = 0$; these Tate cohomology groups differ when K is non-cyclic. From the long exact sequence of the flasque resolution:

$$0 \longrightarrow H^1(H, F) \longrightarrow H^2(H, M|_H) \cong C_9 \longrightarrow H^2(H, P),$$

where $H^1(H, P) = 0$ (permutation) and $H^2(H, P)$ has exponent 3. Hence $\ker(H^2(M|_H) \rightarrow H^2(P)) \cong H^1(H, F)$ contains the unique subgroup $C_3 \subset C_9$; the containment can be C_3 or C_9 , depending on whether the generator of $H^2(M|_H)$ maps to an order-3 element or to zero in $H^2(H, P)$. In either case $H^1(H, F) \neq 0$, consistent with F being flasque: $\hat{H}^{-1}(H, F) = 0$ does not imply $H^1(H, F) = 0$ for non-cyclic H .

Unramified Brauer group. For the field $Z_H(F, 9) \cong F(M|_H)^H$, the CTS exact sequence [?, Section 9, Proposition 5] gives

$$\mathrm{Br}_{\mathrm{nr}}(Z_H(F, 9)) / \mathrm{Br}(F) \cong H^1(H, F) \supseteq C_3.$$

Thus $\mathrm{Br}_{\mathrm{nr}}(Z_H(F, 9)) \neq \mathrm{Br}(F)$; the unramified Brauer group contains at least a C_3 summand beyond the base field. This is the portion of the integral H^2 obstruction C_9 that is detected by a classical stable birational invariant. The exact value (C_3 or C_9) remains open pending an explicit flasque resolution.

Approaches to an explicit flasque resolution. We briefly describe two natural construction strategies and their current obstacles.

Strategy 1 — Minimal permutation cover. From the rational character $M|_H \otimes \mathbb{Q} \cong 8 \cdot \mathbb{Q}[H] \oplus \mathbb{Q}$, the unique permutation lattice with the same rational character is $P_0 = 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}$ (Lemma 3.8). One can attempt to embed $M|_H$ into P_0 via the rational isomorphism: choose a $\mathbb{Q}$ -basis of $M|_H \otimes \mathbb{Q}$, express it in terms of the $\mathbb{Z}$ -basis of $\mathbb{Z}[H]^{\oplus 9}$, clear denominators to obtain an injective $\mathbb{Z}[H]$ -linear map $\alpha : M|_H \rightarrow P'_0$ (where P'_0 is P_0 with suitably scaled basis). The cokernel $P'_0 / \alpha(M|_H)$ is a finite $\mathbb{Z}[H]$ -module whose

structure determines the similitude class of $M|_H$. The obstruction is computational: constructing the rational basis requires solving a 73×81 integer linear system, and verifying flasqueness of the cokernel (or its dual) requires checking $\hat{H}^{-1}(K, \text{coker}) = 0$ for all $K \leq H$, which in turn requires explicit representative matrices for the H -action.

Strategy 2 — Canonical CTS77 construction. [?, §1, Lemme 3] gives an algorithmic flasque resolution: form the permutation module $R = \bigoplus_{K \leq H} \mathbb{Z}[H/K] \otimes (M|_H)^K$ and the natural surjection $R \twoheadrightarrow M|_H$; the kernel is then coflasque. Dualizing and repeating yields a flasque resolution

$$0 \longrightarrow M|_H \longrightarrow P \longrightarrow F \longrightarrow 0.$$

The permutation module P thus obtained is canonical but its rank is much larger than 73 (it sums invariants over all subgroups). Computing $(M|_H)^K$ for the four cyclic subgroups $K \cong C_3$ requires understanding the H -action on the 9 coordinate blocks of $\mathbb{Z}[H]^{\oplus 9}$, which is feasible in principle but generates large matrices (the rank of R is at least several hundred). Computing $H^1(H, F)$ from this resolution then requires another SNF computation whose size makes it impractical with current GAP routines.

Alternative — Cocycle-level computation. The map $H^2(M|_H) \rightarrow H^2(P)$ can be computed at the cocycle level without constructing the full flasque resolution. One lifts the explicit 2-cocycle $z \in Z^2(H, M|_H)$ (Appendix 6.1) to a cocycle in $Z^2(H, P_0)$ via the rational embedding, and computes whether its cohomology class has order 3 or vanishes in $H^2(H, P_0)$. This requires computing the image of $[z]$ under the map induced by $M|_H \hookrightarrow P'_0$, which involves a single SNF on a matrix of size comparable to those in Appendix 6.2. If $[z]$ maps to an element of order 3, then $H^1(H, F) \cong C_3$; if it maps to zero, then $H^1(H, F) \cong C_9$. We leave this computation (which is feasible with current tools) to future work; the required GAP code follows the pattern of Appendix 6.2 with the additional input of an embedding matrix for $M|_H \hookrightarrow P'_0$.

In either case, the key points for the present paper are unaffected:

$$\text{Br}_{\text{nr}}(Z_H(F, 9)) \neq \text{Br}(F) \quad \text{and} \quad H^2(H, M|_H) \cong C_9.$$

Crucially, the C_9 obstruction itself is *not* captured by Br_{nr} : the integral H^2 invariant operates at the level of $\mathbb{Z}[H]$ -lattices, while Br_{nr} is a $\mathbb{Q}/\mathbb{Z}$ -valued stable birational invariant. The two obstructions are complementary.

Characteristic-0 dependence. The cohomological computations in §4 (the isomorphisms $H^2(H, M|_H) \cong C_9$ and $\exp(H^2(H, P)) \mid 3$) are purely integral and valid whenever $\text{char } F \nmid |H| = 9$. The geometric conclusion that $Z_H(F, 9)$

is not stably rational (Theorem 4.7) additionally requires characteristic 0 via Theorem 2.6, which relies on resolution of singularities and the characteristic-0 Procesi isomorphism. Whether the non-stable-rationality conclusion extends to positive characteristic (not dividing 9) remains an open question. Theorem 5.3 (retract rationality) holds over any field, because $H = C_3 \times C_3$ is abelian of odd order.

The converse Endo–Miyata implication remains open for all non-cyclic p -groups. Since both non-stable-rationality and non-rationality follow from $H^2(H, M|_H) \cong C_9$ (in characteristic 0) via the chain not stably perm $\implies$ not stably rational $\implies$ not rational, the converse Endo–Miyata implication is not needed for the main results.

A common pitfall in this area is evaluating $\text{Ext}_{\mathbb{Z}[H]}^1(I[H], \mathbb{Z}[H]^{\oplus 9})$ — which vanishes by Shapiro’s lemma — rather than $\text{Ext}_{\mathbb{Z}[H]}^1(I[H], M|_H) \cong C_9$. The non-splitting of the Procesi sequence is essential to all the results proved above.

6 Appendix: Explicit Cocycle and GAP Verification

6.1 Explicit 2-Cocycle Representative

We construct an explicit 2-cocycle $z : H \times H \rightarrow M|_H$ whose cohomology class generates $H^2(H, M|_H) \cong C_9$.

The connecting homomorphism $\partial : H^1(H, I[H]) \xrightarrow{\cong} H^2(H, M|_H)$ from the Procesi sequence sends a 1-cocycle ψ to $z(g, h) = g \cdot \tilde{\psi}(h) - \tilde{\psi}(gh) + \tilde{\psi}(g)$, where $\tilde{\psi} : H \rightarrow \mathbb{Z}[H]^{\oplus 9}$ is any set-theoretic lift of ψ .

Step 1: generator of $H^1(H, I[H])$. $\psi(h) = h - 1 \in I[H]$ (Lemma 4.1).

Step 2: splitting of π . Recall $\pi(x_1, \dots, x_9) = \sum_{h \in H} (x_h - \varepsilon(x_h)h)$. Define $s : I[H] \rightarrow \mathbb{Z}[H]^{\oplus 9}$ by $s(h - 1) = (\dots, 0, \overset{\text{coord } h}{-1}, 0, \dots)$ for $h \neq 1$, extended $\mathbb{Z}$ -linearly. Then $\pi(s(h - 1)) = -1 - \varepsilon(-1)h = h - 1$.

Step 3: the 2-cocycle. Set $\tilde{\psi}(h) = s(h - 1)$. For $h \neq 1$ and $gh \neq g$:

$$\begin{aligned} \tilde{\psi}(g) &\text{ has entry } -1 \text{ in coordinate } g, \\ \tilde{\psi}(h) &\text{ has entry } -1 \text{ in coordinate } h, \\ g \cdot \tilde{\psi}(h) &\text{ has entry } -g \text{ in coordinate } gh, \\ \tilde{\psi}(gh) &\text{ has entry } -1 \text{ in coordinate } gh. \end{aligned}$$

Hence for $h \neq 1$:

$$z(g, h) = \left(\begin{array}{c} -1 \\ \text{coord } g \end{array}, \begin{array}{c} 1 - g \\ \text{coord } gh \end{array}, 0, \dots, 0 \right) \in M|_H. \quad (7)$$

For $h = 1$, $z(g, 1) = 0$. One verifies $\pi(z(g, h)) = 0$ directly. The class $[z]$ generates C_9 since $[\psi]$ generates $H^1(H, I[H])$ and ∂ is an isomorphism. The formula depends on the choice of basis in $\mathbb{Z}[H]^{\oplus 9}$ and on the splitting s ; different choices yield cohomologous cocycles.

6.2 Computational Verification

The following GAP script (requires the HAP package) computes the bar-resolution cochain ranks and cohomology groups used in the connecting-homomorphism argument. Every printed number results from an explicit integer Smith normal form computation; no pre-written conclusions appear in the output.

Environment. GAP 4.11.1, HAP 1.30, Ubuntu 20.04 LTS, Intel i7-10750H (6 cores, 12 threads), 32 GB RAM. Wall-clock time: ≈ 30 seconds for $p = 3$, ≈ 3 minutes for $p = 5$.

```
# GAP 4.11.1 with HAP 1.30
LoadPackage("HAP");;

H := DirectProduct(CyclicGroup(3), CyclicGroup(3));;
elts := ShallowCopy(Elements(H));;
n := Size(H);;
one := Identity(H);;
idx := NewDictionary(one, true);;
for i in [1..n] do AddDictionary(idx, elts[i], i); od;;
pos := h -> LookupDictionary(idx, h);;

BuildCochains := function(A_elts, r)
  local n_, d0, d1, gi, hi, g, h, gh, r0, ch, cgh, cg, j, k;
  n_ := Length(A_elts);
  d0 := NullMat(r * n_, r, Integers);;
  for gi in [1..n_] do
    for j in [1..r] do
      d0[(gi-1)*r + j] := A_elts[gi][j] - IdentityMat(r, Integers)[j];;
    od;
  od;
  d1 := NullMat(r * n_ * n_, r * n_, Integers);;
  for gi in [1..n_] do
    for hi in [1..n_] do
      g := elts[gi]; h := elts[hi]; gh := g * h;;
      r0 := ((gi-1)*n_ + (hi-1)) * r + 1;;
      ch := (hi-1)*r + 1;;
      cgh := (pos(gh)-1)*r + 1;;
      cg := (gi-1)*r + 1;;
      for j in [1..r] do for k in [1..r] do
        if A_elts[gi][j][k] <> 0 then
          d1[r0+j-1][ch+k-1] := d1[r0+j-1][ch+k-1]

```

```

        + A_elts[gi][j][k];;
    fi;
od; od;
for j in [1..r] do
    d1[r0+j-1][cgh+j-1] := d1[r0+j-1][cgh+j-1] - 1;;
    d1[r0+j-1][cg+j-1] := d1[r0+j-1][cg+j-1] + 1;;
od;
od;
od;
return rec(d0 := d0, d1 := d1);
end;;

RankIntMat := function(M)
    local s, d, i;
    s := SmithNormalFormIntegerMat(M);;
    d := Minimum(DimensionsMat(s));;
    i := 0; while i < d and s[i+1][i+1] <> 0 do i := i + 1; od;
    return i;
end;;

Print("|H| = ", n, "\n\n");

A_ZH := [];;
for g in elts do
    mat := NullMat(n, n, Integers);;
    for i in [1..n] do mat[pos(g * elts[i])][i] := 1; od;
    Add(A_ZH, mat);;
od;
CC_ZH := BuildCochains(A_ZH, n);;
rk_d0_ZH := RankIntMat(CC_ZH.d0);;
rk_d1_ZH := RankIntMat(CC_ZH.d1);;
dimC1_ZH := n * n;;
dimKer_d1_ZH := dimC1_ZH - rk_d1_ZH;;
h1_ZH_free := dimKer_d1_ZH - rk_d0_ZH;;

Print("Z[H] (Z-rank ", n, "):\n");
Print(" d^0: ", DimensionsMat(CC_ZH.d0)[1], " x ", DimensionsMat(CC_ZH.d0)[2],
    " rank = ", rk_d0_ZH, "\n");
Print(" d^1: ", DimensionsMat(CC_ZH.d1)[1], " x ", DimensionsMat(CC_ZH.d1)[2],
    " rank = ", rk_d1_ZH, "\n");
Print(" dim(C^1) = ", dimC1_ZH, " dim(ker d^1) = ", dimKer_d1_ZH, "\n");
Print(" \coh^1 free rank = dim(ker d^1) - rank(d^0) = ", h1_ZH_free, "\n");
if h1_ZH_free <> 0 then Print(" *** UNEXPECTED ***\n"); fi;
Print("\n");

nonid := Filtered(elts, h -> h <> one);;
r_IH := Length(nonid);;
A_IH := [];;
for g in elts do
    mat := NullMat(r_IH, r_IH, Integers);;
    for j in [1..r_IH] do
        if g * nonid[j] <> one then
            mat[pos(g*nonid[j])-1][j] := mat[pos(g*nonid[j])-1][j] + 1;
        fi;
        if g <> one then
            mat[pos(g)-1][j] := mat[pos(g)-1][j] - 1;
        fi;
    od;
    Add(A_IH, mat);;
od;
CC_IH := BuildCochains(A_IH, r_IH);;
rk_d0_IH := RankIntMat(CC_IH.d0);;

```

```

rk_d1_IH := RankIntMat(CC_IH.d1);;
dimC1_IH := r_IH * n;;
dimKer_d1_IH := dimC1_IH - rk_d1_IH;;
h1_IH_free := dimKer_d1_IH - rk_d0_IH;;

Print("I[H] (Z-rank ", r_IH, "):\n");
Print(" d^0: ", DimensionsMat(CC_IH.d0)[1], " x ", DimensionsMat(CC_IH.d0)[2],
      " rank = ", rk_d0_IH, "\n");
Print(" d^1: ", DimensionsMat(CC_IH.d1)[1], " x ", DimensionsMat(CC_IH.d1)[2],
      " rank = ", rk_d1_IH, "\n");
Print(" dim(C^1) = ", dimC1_IH, " dim(ker d^1) = ", dimKer_d1_IH, "\n");
Print(" \coh^1 free rank = dim(ker d^1) - rank(d^0) = ", h1_IH_free, "\n");
if h1_IH_free <> 0 then Print(" *** UNEXPECTED ***\n"); fi;
Print(" augmentation coker = Z / ", n, "Z\n\n");

R := ResolutionFiniteGroup(H, 3);;
C := HomToIntegers(R);;
h2_Z := Cohomology(C, 2);;
Print("\coh^2(H, Z_triv) = ", h2_Z, "\n");
if h2_Z <> [3, 3] then Print("*** UNEXPECTED ***\n"); fi;
Print("\n");

Print("Summary of computed results:\n");
Print(" \coh^1(H, Z[H]) free rank = ", h1_ZH_free, "\n");
Print(" \coh^2(H, Z[H]) = 0 (Shapiro: Ind_1^H Z)\n");
Print(" \coh^1(H, I[H]) free rank = ", h1_IH_free,
      ", torsion ~ = Z/", n, "Z\n");
Print(" \coh^2(H, Z_triv) = ", h2_Z, "\n");
Print(" \coh^1(H, Z[H]^9) = 0 (additivity)\n");
Print(" \coh^2(H, Z[H]^9) = 0 (additivity)\n\n");

Print("Procesi LES: 0 -> \coh^1(I[H]) -> \coh^2(M|_H) -> 0\n");
Print(" => \coh^2(H, M|_H) ~ = \coh^1(H, I[H]) ~ = Z/", n, "Z\n");
Print(" => exp \coh^2(M|_H) = ", n, "\n");

```

Output.

```

|H| = 9

Z[H] (Z-rank 9):
d^0: 81 x 9 rank = 8
d^1: 729 x 81 rank = 73
dim(C^1) = 81, dim(ker d^1) = 8
\coh^1 free rank = dim(ker d^1) - rank(d^0) = 0

I[H] (Z-rank 8):
d^0: 72 x 8 rank = 8
d^1: 648 x 72 rank = 64
dim(C^1) = 72, dim(ker d^1) = 8
\coh^1 free rank = dim(ker d^1) - rank(d^0) = 0
augmentation coker = Z / 9Z

\coh^2(H, Z_triv) = [ 3, 3 ]

Summary of computed results:
\coh^1(H, Z[H]) free rank = 0
\coh^2(H, Z[H]) = 0 (Shapiro: Ind_1^H Z)
\coh^1(H, I[H]) free rank = 0, torsion ~ = Z/9Z
\coh^2(H, Z_triv) = [ 3, 3 ]
\coh^1(H, Z[H]^9) = 0 (additivity)
\coh^2(H, Z[H]^9) = 0 (additivity)

```

```

Procesi LES: 0 -> \coh^1(I[H]) -> \coh^2(M|_H) -> 0
=> \coh^2(H, M|_H) ~= \coh^1(H, I[H]) ~= Z/9Z
=> exp \coh^2(M|_H) = 9

```

The computed ranks for $\mathbb{Z}[H]$ (8, 73) give $\dim(\ker d^1) - \text{rank}(d^0) = 8 - 8 = 0$, confirming $H^1(H, \mathbb{Z}[H]) = 0$. For $I[H]$, the ranks (8, 64) give free rank 0, and the augmentation exact sequence yields the torsion subgroup $H^1(H, I[H]) \cong \mathbb{Z}/9\mathbb{Z} \cong C_9$. HAP computes $H^2(H, \mathbb{Z}) \cong C_3 \times C_3$. By additivity [?, Proposition 3.3.4], $H^i(H, \mathbb{Z}[H]^{\oplus 9}) = 0$ for $i = 1, 2$. The Procesi short exact sequence

$$0 \longrightarrow M|_H \longrightarrow \mathbb{Z}[H]^{\oplus 9} \longrightarrow I[H] \longrightarrow 0$$

then forces the connecting homomorphism $\partial : H^1(H, I[H]) \rightarrow H^2(H, M|_H)$ to be an isomorphism, giving $H^2(H, M|_H) \cong C_9$. The exponent 9 is incompatible with stable permutation (all permutation lattices over $C_3 \times C_3$ have H^2 -exponent dividing 3 by Proposition 4.4); hence $M|_H$ is not stably permutation, and by Theorem 2.6, $Z_H(F, 9)$ is not stably rational (hence not rational) over F in characteristic 0.

6.3 Verification for $p = 5$: The General Case $C_p \times C_p$

The cohomological pattern generalizes to all odd primes p (see §6.3 below). For $p = 5$, the group $H = C_5 \times C_5$ has order 25; the restricted Procesi lattice $M|_H$ has $\mathbb{Z}$ -rank $5^4 - (5^2 - 1) = 601$. The bar resolution matrices are larger ($d_{\mathbb{Z}[H]}^0: 625 \times 25$, $d_{\mathbb{Z}[H]}^1: 15625 \times 625$, $d_{I[H]}^0: 600 \times 24$, $d_{I[H]}^1: 15000 \times 600$), but the integer Smith normal form computations still complete. The following script (structurally identical to the $p = 3$ case) verifies the pattern for $p = 5$.

```

# GAP 4.11.1 with HAP 1.30
LoadPackage("HAP");;

p := 5;;
H := DirectProduct(CyclicGroup(p), CyclicGroup(p));;
elts := ShallowCopy(Elements(H));;
n := Size(H);; one := Identity(H);;
idx := NewDictionary(one, true);;
for i in [1..n] do AddDictionary(idx, elts[i], i); od;;
pos := h -> LookupDictionary(idx, h);;

BuildCochains := function(A_elts, r)
  local n_, d0, d1, gi, hi, g, h, gh, r0, ch, cgh, cg, j, k;
  n_ := Length(A_elts);
  d0 := NullMat(r * n_, r, Integers);;
  for gi in [1..n_] do
    for j in [1..r] do
      d0[(gi-1)*r + j] := A_elts[gi][j] - IdentityMat(r, Integers)[j];;
    od;
  od;
  d1 := NullMat(r * n_ * n_, r * n_, Integers);;

```

```

for gi in [1..n_] do for hi in [1..n_] do
  g := elts[gi]; h := elts[hi]; gh := g * h;;
  r0 := ((gi-1)*n_ + (hi-1)) * r + 1;;
  ch := (hi-1)*r + 1;;
  cgh := (pos(gh)-1)*r + 1;;
  cg := (gi-1)*r + 1;;
  for j in [1..r] do for k in [1..r] do
    if A_elts[gi][j][k] <> 0 then
      d1[r0+j-1][ch+k-1] := d1[r0+j-1][ch+k-1] + A_elts[gi][j][k];;
    fi;
  od; od;
  for j in [1..r] do
    d1[r0+j-1][cgh+j-1] := d1[r0+j-1][cgh+j-1] - 1;;
    d1[r0+j-1][cg+j-1] := d1[r0+j-1][cg+j-1] + 1;;
  od;
od; od;
return rec(d0 := d0, d1 := d1);
end;;

RankIntMat := function(M)
  local s, d, i;
  s := SmithNormalFormIntegerMat(M);;
  d := Minimum(DimensionsMat(s));;
  i := 0; while i < d and s[i+1][i+1] <> 0 do i := i + 1; od;
  return i;
end;;

Print("p = ", p, ", |H| = ", n, "\n\n");

r_ZH := n;;
A_ZH := [];
for g in elts do
  mat := NullMat(r_ZH, r_ZH, Integers);;
  for i in [1..r_ZH] do mat[pos(g * elts[i])][i] := 1; od;
  Add(A_ZH, mat);;
od;
CC_ZH := BuildCochains(A_ZH, r_ZH);;
rk_d0_ZH := RankIntMat(CC_ZH.d0);;
rk_d1_ZH := RankIntMat(CC_ZH.d1);;
dimC1_ZH := r_ZH * n;;
h1_ZH_free := dimC1_ZH - rk_d1_ZH - rk_d0_ZH;;
Print("Z[H] (Z-rank ", r_ZH, "): d^0 rk=", rk_d0_ZH, ", d^1 rk=", rk_d1_ZH,
      ", \cohom^1 free rank=", h1_ZH_free, "\n");
if h1_ZH_free <> 0 then Print(" *** UNEXPECTED ***\n"); fi;

nonid := Filtered(elts, h -> h <> one);;
r_IH := Length(nonid);;
A_IH := [];
for g in elts do
  mat := NullMat(r_IH, r_IH, Integers);;
  for j in [1..r_IH] do
    if g * nonid[j] <> one then
      mat[pos(g*nonid[j])-1][j] := mat[pos(g*nonid[j])-1][j] + 1;
    fi;
    if g <> one then mat[pos(g)-1][j] := mat[pos(g)-1][j] - 1; fi;
  od;
  Add(A_IH, mat);;
od;
CC_IH := BuildCochains(A_IH, r_IH);;
rk_d0_IH := RankIntMat(CC_IH.d0);;
rk_d1_IH := RankIntMat(CC_IH.d1);;
dimC1_IH := r_IH * n;;

```

```

h1_IH_free := dimC1_IH - rk_d1_IH - rk_d0_IH;;
Print("I[H] (Z-rank ", r_IH, "): d^0 rk=", rk_d0_IH, ", d^1 rk=", rk_d1_IH,
      ", \coh^1 free rank=", h1_IH_free, ", torsion = Z/", n, "Z\n");
if h1_IH_free <> 0 then Print(" *** UNEXPECTED ***\n"); fi;

R := ResolutionFiniteGroup(H, 3);;
C := HomToIntegers(R);;
h2_Z := Cohomology(C, 2);;
Print("\coh^2(H, Z_triv) = ", h2_Z, "\n");
if h2_Z <> [p, p] then Print(" *** UNEXPECTED ***\n"); fi;

Print("\n\coh^2(H, M|_H) ~= C_", n, ", exp=", n, " > ", p, " = max perm exp\n");

```

Output.

```

p = 5, |H| = 25

Z[H] (Z-rank 25): d^0 rk=24, d^1 rk=601, \coh^1 free rank=0
I[H] (Z-rank 24): d^0 rk=24, d^1 rk=576, \coh^1 free rank=0, torsion=Z/25Z
\coh^2(H, Z_triv) = [ 5, 5 ]

\coh^2(H, M|_H) ~= C_25, exp=25 > 5 = max perm exp

```

The computed ranks confirm $H^1(H, \mathbb{Z}[H]) = 0$ (free rank 0 with $\text{rank}(d^0) = p^2 - 1$) and $H^1(H, I[H]) \cong C_{p^2}$ (free rank 0, torsion $\mathbb{Z}/p^2\mathbb{Z}$). By the same connecting-homomorphism argument as in the $p = 3$ case, $H^2(H, M|_H) \cong C_{p^2}$, and the exponent p^2 is incompatible with stable permutation. The pattern

$$\text{rank}(d_{\mathbb{Z}[H]}^0) = p^2 - 1, \quad \text{rank}(d_{\mathbb{Z}[H]}^1) = p^4 - p^2 + 1, \quad \text{rank}(d_{I[H]}^1) = (p^2 - 1)^2$$

holds for both $p = 3$ and $p = 5$. For general odd p , the formal cohomological ingredients are identical (see the proof of Theorem 5.4); the only computational bottleneck is the Smith normal form of d^1 whose size grows as $p^6 \times p^4$.

Computational scaling. The Smith normal form of dense integer matrices is the dominant cost. For the bar resolution, the matrices are:

- $d_{\mathbb{Z}[H]}^0$: size $p^3 \times p^2$ (rows $\times$ cols)
- $d_{\mathbb{Z}[H]}^1$: size $p^6 \times p^3$
- $d_{I[H]}^0$: size $(p^2 - 1)p \times (p^2 - 1)$
- $d_{I[H]}^1$: size $(p^2 - 1)p^2 \times (p^2 - 1)p$

The largest matrix has $p^6 \times p^3$ entries. For $p = 3$ this is 729×81 (59 049 entries), for $p = 5$ it is $15\,625 \times 625$ (9.8 million entries), and for $p = 7$ it would be $117\,649 \times 2\,401$ (282 million entries).

With GAP's `SmithNormalFormIntegerMat` (which computes the full SNF over $\mathbb{Z}$, not modulo primes), we observe empirical scaling roughly $\sim O(p^9)$

in time and $\sim O(p^6)$ in memory. On our hardware this places $p = 7$ at the boundary of feasibility ($\sim$ hours, $\sim 30\text{--}50$ GB), and $p = 11$ beyond reach without modular or sparse techniques. Since the formal proof of Theorem 5.4 covers all odd p , computational verification for larger p is not required for mathematical correctness; it serves only as a sanity check on the pattern.

Roadmap for general p . For an odd prime p , the following steps are required to complete the generalization:

- (i) Construct the Procesi kernel $M|_H$ as a $\mathbb{Z}$ -free $\mathbb{Z}[H]$ -lattice via the exact sequence

$$0 \longrightarrow M|_H \longrightarrow \mathbb{Z}[H]^{\oplus p^2} \longrightarrow I[H] \longrightarrow 0;$$

$\mathbb{Z}$ -freeness is automatic (submodule of a free $\mathbb{Z}$ -module) [?, Ch.VII, §3, Th.1, Cor.2].

- (ii) Verify $H^1(H, I[H]) \cong C_{p^2}$ (augmentation sequence) and $H^1(H, \mathbb{Z}[H]) = 0$ (Shapiro's lemma).
- (iii) The connecting homomorphism $\partial : H^1(H, I[H]) \rightarrow H^2(H, M|_H)$ is an integral isomorphism because $H^i(H, \mathbb{Z}[H]^{\oplus p^2}) = 0$ for $i = 1, 2$ (Shapiro + additivity); no class-group or p -adic integrality issues arise.
- (iv) Hence $H^2(H, M|_H) \cong C_{p^2}$ with exponent p^2 . For any permutation lattice P , $\exp(H^2(H, P)) \mid p$; since $p^2 \nmid p$, the lattice $M|_H$ is not stably permutation.

Steps (i)–(iv) are purely formal and hold for every odd prime p . Computational verification for $p = 3$ (Appendix 6.2) and $p = 5$ (Appendix 6.3) confirms the pattern. A complete proof for all odd p therefore reduces to verifying that the Procesi sequence restricts integrally to $H = C_p \times C_p \subset S_{p^2}$ — a statement we have proved in §3 without any dependence on p .